

Hotel Booking Cancellation Prediction: An Early Risk Assessment Approach

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We propose to estimate the probability that a hotel booking will eventually cancel, using information intended to be available when the booking is created. The central question is whether booking characteristics improve probability estimates beyond the historical cancellation rate. This is a preliminary study: source and schema checks are complete, but exploratory analysis and model fitting remain planned.

1 Motivation and potential use

A booking reserves capacity without guaranteeing that the guest will arrive. Cancellations can leave rooms unsold and make expected demand harder to estimate. This problem connects a clear business decision to a measurable prediction task. A useful model could support forecasts of retained bookings and help staff prioritize voluntary reconfirmation. The project will evaluate predictive quality, not claim a measured revenue gain.

2 Prediction target and loss

One observation is one booking. Y is the dataset label `is_canceled`: 1 for a canceled booking and 0 otherwise. We retain the published label and will check how it aligns with Canceled, Check-Out, and No-Show statuses before interpreting it. X contains a restricted set of candidate booking characteristics. The output $p(X)$ estimates $P(Y = 1 | X)$, rather than only a yes/no decision.

The primary evaluation criterion is mean binary Log Loss, also called cross-entropy:

$$L = -(1/n) \sum_i [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)].$$

Lower values are better. Confident errors receive a larger penalty, which suits a task in which the probability itself matters. Evaluation will clip probabilities to $[10^{-15}, 1 - 10^{-15}]$ for numerical stability. This predictive loss does not directly represent the hotel's monetary cost.

3 Related work

Antonio, de Almeida, and Nunes (2017) studied cancellation classification using four resort hotels and discussed its role in demand management [2]. Their study establishes a relevant application, but its reported performance is not a benchmark for our feature restrictions or split. Their 2019 data article documents the public booking records and extraction process [1]. Our project is a modest comparison of probability baselines, logistic regression, and random forest, with particular attention to prediction timing.

4 Data source and structure

The original data article reports two Portuguese hotels with bookings scheduled to arrive from July 1, 2015 through August 31, 2017. The resort dataset has 40,060 observations and the city dataset has 79,330, each with 31 variables [1]. These are two individual properties, not representative samples of all resort and city hotels.

We selected the publicly accessible TidyTuesday hotels.csv as a reproducible merged copy [3]. A direct schema check on October 8, 2026 confirmed 119,390 rows and 32 columns, including the target. The extra hotel column identifies the source property. Thus, 32 columns describes this common merged format, not each original dataset or the final number of predictors. No cancellation rates, missing-value counts, or model scores have been computed for this proposal.

Data element	Examples and interpretation
Numeric and count fields	lead_time; adults, children, babies; planned weekday/weekend nights
Categorical fields	hotel, meal, market_segment, distribution_channel, reserved_room_type
Calendar fields	Arrival year, month, week number, day of month; combine into an intended arrival date
Identifiers	agent and company are category codes, not quantities
Outcome fields	is_canceled is Y; reservation_status and reservation_status_date are excluded from X

5 Information available at prediction time

The intended prediction time is initial booking creation. The public file does not provide a complete history of booking-time snapshots. The original authors used change-log values relative to the day before arrival when available [1]. A value that is known before arrival may still be unavailable at initial booking. Removing the obvious outcome fields alone cannot establish a valid early prediction setting.

The provisional core includes hotel, lead_time, intended arrival season, planned stay length, party size, reserved_room_type, meal, market_segment, distribution_channel, and customer_type. Their recorded values remain subject to a timing audit because bookings can change. Guest-history fields may be added only if they refer strictly to outcomes known before the current booking.

The default early model will exclude assigned_room_type, booking_changes, days_in_waiting_list, adr, deposit_type, parking requirements, and total_of_special_requests. Room assignment and accumulated changes can occur later. ADR and deposits are not guaranteed to describe the initial quoted price or policy, and requests can change. A sensitivity model may add documented initial values. If initial values cannot be verified, results will be described as an offline approximation, with that limitation carried into the conclusion.

6 Data quality and leakage audit

Before training, we will preserve the raw file, record its checksum, and produce a field-level audit of types, null encodings, plausibility, and availability. Source definitions guide the checks below, but they are not a substitute for inspecting this exact file.

Issue	Planned handling
Missingness and meaning	Distinguish unknown values from “not applicable.” In particular, a missing agent or company may mean no intermediary. Do not replace all missing codes with numeric zero.
Mixed types	Treat identifiers as categorical. Parse calendar fields explicitly and check category spelling. Learn numeric imputation and category handling from training data only.
Implausible values	Flag negative lead times, invalid dates, unusual stay lengths, and implausible party sizes. Review their meanings before filtering. Log every exclusion.
Repeated rows	Count exact duplicate rows and inspect patterns. Anonymous bookings can legitimately share all fields. Do not assume every identical row is a duplicate person or remove it automatically.
Outcome leakage	Remove reservation_status and reservation_status_date from predictors and derived features. They describe final status and its timing, directly revealing or following the outcome.
History and updates	Check guest-history timing and mutable reservation attributes. Reject fields that use events occurring after the intended prediction time.

Exploratory data analysis will report cancellation frequency by hotel and time period, missingness by field, lead-time distributions, and category coverage. Associations will be treated as descriptive rather than causal. The test set will remain untouched during feature selection, cleaning-rule selection, and tuning.

Expected challenges

Potential class imbalance makes accuracy alone inadequate. High-cardinality identifiers and rare categories can overfit. Seasonality and changing hotel policies may cause distribution shift. The anonymous release prevents reliable grouping of repeat customers or linked group bookings, so some dependence may remain across subsets. Two properties observed in 2015–2017 offer limited evidence for a different hotel or for current operating conditions.

The main unresolved risk is temporal validity. The final report will distinguish verified initial fields from assumptions and exclusions. A high predictive score cannot resolve missing booking-time snapshots.

7 Baseline and model training plan

The primary baseline predicts the training-set cancellation rate for every booking: $p_0 = (\sum y_i)/n_{\text{train}}$. It requires no booking features and is the best constant probability for training Log Loss. We will estimate it separately inside each training fold and freeze it for validation or test. An always-not-canceled rule may serve as a secondary accuracy reference, but a hard probability of zero is unsuitable for Log Loss when cancellations occur.

Model	Purpose and planned controls
Logistic regression (LR)	An interpretable probability model with additive effects on log-odds. Median-impute numeric fields, add missingness indicators where appropriate, standardize numeric fields, and one-hot encode categories. Tune L2 regularization strength.
Random forest (RF)	A comparison that can capture nonlinear effects and interactions. Use consistent feature eligibility and categorical encoding. Tune depth, minimum leaf size, and feature subsampling with a small prespecified search. Leaf size matters for stable probability estimates.

Both models will use identical eligible features and held-out subsets. All learned preprocessing will sit inside the training pipeline. We will initially retain the observed class distribution because resampling and class weighting can alter probabilities. If later used, their effect on calibration must be checked. LR optimizes regularized logistic loss. RF grows classification trees, typically using impurity criteria, and averages tree probabilities. Both will be selected by held-out Log Loss.

8 Temporal evaluation design

The proposed primary split uses intended arrival date: July 2015–December 2016 for training, January–March 2017 for validation, and April–August 2017 for a final held-out test. We will report actual subset sizes after the audit. Expanding-window checks within the training period will assess stability before the validation choice. The final test will be opened once after the model, features, and any calibration rule are fixed.

This split evaluates later arrival cohorts. It does not ensure that every test booking was created after training: long-lead bookings can overlap in booking time. It also does not guarantee that training outcomes were known at a historical cutoff. We will inspect the approximate entry date (arrival date minus lead_time) and outcome timing for overlap and label maturity. reservation_status_date may support an audit, but will never enter X.

A genuine deployment simulation would need trustworthy booking-creation timestamps, initial snapshots, and labels available at each training cutoff. Derived dates cannot recover missing snapshots. We will use them only as a sensitivity check. A stratified random split is a secondary diagnostic. Differences between split results, and the incomplete seasonal coverage of the final test, will qualify the conclusion.

9 Evaluation and decision criteria

We will report mean Log Loss as the primary metric, together with Brier score and a reliability plot to assess probability calibration. ROC-AUC measures ranking quality. Precision, recall, F1, and a confusion matrix at a validation-selected threshold will explain classification tradeoffs. No threshold will be selected on the final test, and accuracy will remain a secondary measure.

Results will show each model beside the constant-probability baseline, overall and separately for the city and resort properties. We will also inspect calendar periods and lead-time groups, reporting their sample sizes. Paired bootstrap intervals grouped by arrival week will summarize uncertainty in the difference in test Log Loss, while acknowledging that anonymous linked bookings may violate independence assumptions. If calibration is needed, it will be fitted using training-period out-of-fold predictions, with its method chosen on validation data.

Evidence of useful prediction would be lower test Log Loss than the constant baseline, with reasonably reliable probabilities across both properties and no dependence on excluded fields. A null or unstable improvement is still informative: it would show that the restricted data or the model does not support the intended use. We will not choose a numerical success target before observing the baseline or present planned metrics as results.

10 Impact and next steps

For a set of bookings, $\sum(1 - p_i)$ estimates the expected number retained if probabilities are accurate. Translating this into room occupancy requires stay dates and room quantities. Prediction alone does not show that contacting guests reduces cancellations or that overbooking increases profit. Those decisions require action costs, capacity constraints, and a separate intervention evaluation.

For the October 12 preliminary submission, this proposal and the matching presentation specify the question, verified data structure, and evaluation plan. The next work is to complete the availability and quality audit, run EDA, fit the baseline and LR/RF pipelines, and compare results under the proposed splits. The final presentation will include reproducible code and documented limitations. Individual confidential contribution reports must reflect actual work and are separate from these team materials.

References

- [1] Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. *Data in Brief*, 22, 41–49. <https://doi.org/10.1016/j.dib.2018.11.126>
- [2] Antonio, N., de Almeida, A., & Nunes, L. (2017). Predicting hotel booking cancellations to decrease uncertainty and increase revenue. *Tourism & Management Studies*, 13(2), 25–39. <https://doi.org/10.18089/tms.2017.13203>
- [3] TidyTuesday. (2020, February 11). *Hotels: data dictionary, merge script, and hotels.csv*. <https://github.com/rfordatascience/tidytuesday/tree/main/data/2020/2020-02-11> (accessed October 8, 2026).